

Improving Mechanical Performance for Extreme Environments

Project Description

WSU Mechanics and Materials Laboratory



WSUM-Python Apps

Nelson Goosman

Kenan Anderson

Nicholas Lopez

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I. Introduction

Protection against high velocity projectiles is crucial for both the space and defense industries. For both industries they rely on that protection for not only human safety but also equipment safety. It is imperative to accurately predict when a protective shield will break and how it will break under extreme circumstances.

This is exactly what the goal of our team is, we want to further enhance the program given, so that it can accurately analyze images of these protective shields under extreme and high stress environments. When analyzing these images, we collect data that is crucial to researchers to help figure out why these ceramics and alloys break and how they can improve on them for better overall performance. Currently, the program utilizes image segmentation to format images for analysis. Our team aims to extend the application by incorporating multiple image analysis techniques, batch segmentation capabilities, advanced statistical analysis methods, and a machine learning approach to predict material properties based on specific parameters.

II. Background and Related Work

This project focuses on the automated analysis of microscopic images to enhance material design for extreme or high-stress environments. Specifically, the analysis will include impurity detection in materials and fragmentation assessment following exposure to high-impact forces or extreme temperatures. This research extends the work initiated by a previous capstone group, building upon existing methodologies to further refine material characterization techniques.

Current state-of-the-art (SOTA) methodologies for image segmentation in this domain include both machine learning-based and algorithmic approaches. Deep learning models, particularly convolutional neural networks (CNNs), have demonstrated strong performance in image segmentation tasks. One notable model is U-Net, which was originally developed for biomedical image segmentation [1]. Given its capability for pixel-level classification and its ability to process images at their original resolution, U-Net may prove beneficial for segmenting irregular material structures that differ from conventional real-world imagery, such as people or objects, on which many other SOTA models are trained. Another relevant deep learning approach is Meta's Segment Anything Model (SAM), which is designed for generalizable segmentation across diverse datasets [2].

In addition to machine learning-based methods, algorithmic segmentation techniques provide valuable alternatives. Several approaches will be explored and implemented to allow flexibility in analysis. Otsu's Thresholding, a widely used method for automated crack detection in materials, offers a robust means of analyzing material deformation across large datasets [3]. The Watershed Algorithm, already integrated into this project, segments images based on variations in brightness and contrast, making it a useful tool for delineating material structures [4]. Additionally, Felzenszwalb's efficient graph-based segmentation method presents a comparable alternative to the Watershed Algorithm, offering further customization for users seeking to balance segmentation accuracy and computational efficiency. By incorporating these diverse

segmentation methodologies, the project aims to provide users with a versatile toolkit, enabling them to optimize their analyses based on specific requirements.

To achieve a successful outcome, a comprehensive understanding of several key areas is necessary. The team will need to develop expertise in image processing techniques for analyzing electron microscopy images, both conceptually and through Python-based implementations. Additionally, proficiency in statistical analysis methods will be essential to extract and convey meaningful insights to end users. The project also requires the development of a production-level application capable of seamless deployment across various systems, necessitating an understanding of environment management for streamlined installation. Furthermore, improving proficiency in graphical user interface (GUI) design will be critical to ensuring an intuitive and accessible user experience.

By integrating advanced segmentation techniques with robust computational tools, this project seeks to enhance the efficiency and accuracy of automated material analysis, ultimately contributing to improved material design for high-stress applications.

III. Project Overview

The Multiscale Mechanics and Materials Laboratory at the School of Mechanical and Materials Engineering from Washington State University is looking to develop a program for an automated analysis of microscopic images from various material impacts. These materials, such as boron carbide ceramics and niobium-titanium alloys, are used and designed to withstand high-velocity projectile impacts. Dr. Zare's laboratory utilizes characterization and data science to predict and improve the mechanical performance of materials for the extreme dynamic environments found in space travel, aerospace, and defense collisions.

Previous experiments have created impact velocities ranging from 1 to 4.5 km/s, after which microscopic images of the resulting fractures and void spaces were collected. These images contain critical information about how the tested material fails under extreme conditions. The overarching project goal is to create an advanced computational tool to automate the segmentation and statistical analysis of these images. This allows for a better understanding of the materials' structural integrity post-impact. Additionally, this project aims to integrate machine learning to predict material properties based on derived metrics and a provided set of material parameters.

The resultant project will meet the following high-level goals and outcomes:

- An improved, existing program to support multi-image analysis
- An automated system for analyzing microstructural images of high-impact collisions
- Identification of optimal impact conditions
- Additional and optional segmentation/statistical analyses methods
- A trained machine learning model capable of predicting material performance
- A predictive framework utilizing training data and material parameters to design resilient materials for aerospace, defense, and space travel applications.

- A documented, reproducible, and extendable framework for further research in materials science and impact engineering

The goals and desired outcomes for this project are clearly defined and understood by the mentor and project team. We intend to extend the functionality of a previously developed image segmentation and statistical analysis program to support batch processing, allowing multiple images to be analyzed simultaneously. Similarly, we will integrate additional image segmentation algorithms to improve the accuracy of feature identification in distinguishing void spaces and crack patterns, as well as generalizing the program to fit the unique needs of different researchers. With advanced segmentation implemented, we can better extract quantitative metrics such as void size, shape, aspect ratio, and spatial distribution. In correlation with mechanical impact test data, these statistics will begin to determine failure patterns. In efforts to supplement the core algorithmic approach, we aim to train a machine learning model to predict material properties based on extracted image features and mechanical testing data. In developing a framework for a backward pass analysis, material modifications will be suggested to achieve desired mechanical properties. At a high level, we will implement a streamlined pipeline that automates data collection, segmentation, statistical analysis, and predictive machine learning.

This project reflects a significant step towards integrating cutting-edge computational techniques with specialized materials science research. By leveraging image segmentation, statistical analysis, and machine learning, we aim to provide insights into material failure and predictive tools for designing new and advanced materials. Our team will build upon prior work, refine the existing approach, and introduce new methodologies to improve the accuracy, efficiency, and applicability of the program.

IV. Client and Stakeholder Identification and Preferences

Our client is Dr. Zare and the multiscale mechanics and materials laboratory at Washington State University. Now there are no stakeholders, but in the near future we could see stakeholders and other clients that would use this application.

The current need of our client is to improve on the existing application and continue working on innovative ways to help accurately gather data of these protective shields under harsh environments and identify the imperfections, to boost the efficiency of protective shields. Our client also wants us to make the application a highly generalized tool that can be used beyond the current scope to benefit researchers.

In the near future, this application can benefit a larger range of clients and stakeholders, clients can range from anyone in defense contracting, space agencies and the department of defense. Our stakeholders could range from researchers, engineers, and academic institutions. They would all benefit from this project, creating a new and efficient way for stakeholders and clients alike to analyze and digest information on materials after they've been subjected to harsh environments.

V. Glossary

SAM: Segment Anything Model

SOTA: State-of-the-art

GUI: Graphical user interface

CNN: Convolutional neural network

Image Segmentation: a technique that partitions a digital image into discrete groups of image segments.

Batch Segmentation: the process of dividing a large dataset of customer or user data into different segments based on specific criteria

VI. References

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